

Name: Thaneshwar M

Roll No: 240701561

UID EX 10

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Inventory Management Visualization</title>
7  <style>
8      body {
9          font-family: Arial, sans-serif;
10         text-align: center;
11         margin: 50px;
12     }
13     canvas {
14         margin: 20px auto;
15     }
16 </style>
17 </head>
18 <body>
19     <h1>Inventory Management System</h1>
20     <canvas id="pieChart" width="400" height="400"></canvas>
21     <canvas id="barChart" width="400" height="400"></canvas>
22     <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
23     <script src="script.js"></script>
24 </body>
25 </html>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

1  const inventoryData = {
2    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
3    datasets: [{label: 'Items in Stock',data: [200, 150, 100, 80, 50],backgroundColor: ['#FF6384','#36A2EB','#FFCE56','#4BC0C0','#9966FF'],}]
4  };
5
6
7  const ctxPie = document.getElementById('pieChart').getContext('2d');
8  const pieChart = new Chart(ctxPie, {
9    type: 'pie',
10   data: inventoryData,
11   options: {
12     responsive: true,
13     title: {
14       display: true,
15       text: 'Inventory Distribution'
16     }
17   }
18 });
19
20
21 const ctxBar = document.getElementById('barChart').getContext('2d');
22 const barChart = new Chart(ctxBar, {
23   type: 'bar',
24   data: inventoryData,
25   options: {
26     responsive: true,
27     title: {
28       display: true,
29       text: 'Items in Stock by Category'
30     },
31   },
32   scales: {
33     yAxes: [{
34       ticks: {
35         beginAtZero: true
36       }
37     }]
38   }
39 });
40 });

```

OUTPUT:

Inventory Management System

